

Genomic Biology

Review

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Introduction

Genomics is changing our understanding of biology. At present, the greatest impact of genomic research has come from DNA sequencing projects. These have revealed the genetic complement of yeast, worms, and flies, and more than 20 species of bacteria and provided access to the encoded functions of these organisms. Analysis of genomes has also provided insights into polymorphisms within species, protein interaction, and evolution. Other genomic (defined as high-throughput, not necessarily hypothesis-dependent) methods that examine mRNA and proteins will offer insights into mRNA expression, protein expression, protein localization, and protein interactions and may cast light on the flow of information within signaling pathways. The volume of genomic information has prompted the development of general experimental methods that allow quick tests of the hypotheses it suggests. Insights from genomic biology will greatly affect medicine and agriculture. Genomics is also changing the biological community. At its beginning, biology involved observing nature and experimenting on isolated parts of it. Genomic research now generates new types of complex observational data derived from nature. For now, a time when much of the data are closely linked to observation and when some researchers have ceased to perform hypotheses-driven research, researchers still need to practice critical thought. In the future, to make sense of the data, genomic researchers will need to partly recapitulate the development of biology itself, by devising new ways to handle and isolate appropriate subsets of the information and new heuristics for reasoning from that information and suggesting further experiment. Genomics may also lead to a more radical development. As the genomic inventories approach closure, the mass of this data will spur attempts to devise computational frameworks that integrate biological knowledge about cellular components and attempt to predict system behavior. During the early twenty-first century, this more predictive biology will have positive consequences for health and agriculture and will speed the development of a design-based biological engineering of cells and organisms to perform new functions.

From the Beginning to the Present Some Perspective

Biology arose from the study of nature and from performing experiments on isolated parts of it. Genomic biology provides new kinds of data that are derived from nature and are nature-like in their complexity. During the late 1980s, genomics referred to the generation and analysis of information about genes and genomes, so

long as that information could be produced systematically, by DNA sequencing projects, which were resources specialized for that purpose. During the middle 1990s, a new term, functional genomics, came to refer to generation and analysis of the information about what genes do, so long as the information was produced systematically. Since then, a number of other terms have been introduced: proteomics, transcriptomics, etc.; in fact, Weinstein (1998) has suggested that all such work be termed “omics.” Whatever the merits of that idea, here I will use genomics in a broad sense, to mean the generation of information about living things by systematic approaches that can be performed on an industrial scale.

Information from Gene Mapping and Sequencing

The wealth of information coming from sequencing projects for bacteria, yeast, flies, mice, and humans has been reviewed extensively elsewhere (*Science*, October 15, 1999, and references therein). However, no review on genomics would be complete without mention of the following general findings.

Linkage Information. The finding (Kan and Dozy, 1978) and elaboration into doctrine (Botstein et al., 1980) that DNA polymorphisms (then, restriction fragment linkage polymorphisms, or RFLPs; now, single nucleotide polymorphisms, or SNPs) could be followed as Mendelian alleles in pedigrees led to a great expansion of the density of genetic maps in all organisms so studied. The ability to find such systematically and unambiguously scorable markers linked to inherited traits allowed isolation of genes responsible for these traits (the first being for that for Huntington's chorea [Gusella et al., 1983] by DNA walking [Bender et al., 1983]). Most human medical genetics still depends on isolation of individual disease genes by this tactic.

Information about Genome Organization. Sequencing a genome reveals the organization of its contents. Perhaps surprisingly, even a first order inspection of this organization is often interesting, revealing facts whose functional significance is sometimes unclear but that likely reflect some kind of previous selection rather than evolutionary accident. For example, consider *Saccharomyces cerevisiae*. The genome shows evidence of two primordial doublings (Wolfe and Shields, 1997; Keogh et al., 1998) that resulted in a tetraploid genome from which genes were subsequently lost. In this genome, redundant functions are conserved near the telomeres, while essential genes are not found there (Winzeler et al., 1999). In *S. cerevisiae*, small chromosomes have higher recombination rates than larger ones, and ORFs with higher than average CG content at silent positions tend to be on those smaller chromosomes (Bradnam et al., 1999). AT-rich areas near centromeres and telomeres show lower rates of meiotic recombination than the rest of the genome (Goffeau et al., 1996). ORF density falls off near telomeres (Bradnam et al., 1999). All fascinating, but so far uninterpretable.

Information about Protein Complement. The combination of sequences from genomic DNA, EST, and full-length cDNA has resulted in increasingly complete lists

of encoded effector molecules. Having more genomes gets one more information, in the sense that knowledge of a gene's function in one organism suggests functions for genes of similar sequence in others. In fact, at least 10% of the papers in *Cell* in the past 15 years have rested on this fact, and the inventors of the sequence comparison algorithms FASTA (Pearson and Lipman, 1988) and BLAST (Altschul et al., 1990) may thus arguably share credit with the cited authors for many biological facts uncovered during the past decade. Of course, absent sequence similarity to a known protein, neither the biochemical nor the higher order function of a protein is immediately apparent. Thus, at present, knowledge of the function of genomic protein complements remains murky. The murk is deepened by the fact that most sequences are of bacterial genomes and that most analysts who study these sequences are more knowledgeable about microbial metabolism than other aspects of biology. In consequence, the analysis of whole protein complements has been most useful in understanding intermediate metabolism, and, for bacteria and sequenced eukaryotes, most of the players in basic anabolism and catabolism are revealed. Beyond metabolism, the general pictures of biological function revealed by coding sequences are evocative but not conclusive: bacteria have more transporters than we had imagined, and worms have more ion channels than they have nerve cells. At present, the most important observation from genome sequence is probably that, depending on one's criteria, the function of between 15% and 40% of the proteins encoded by any genome is not apparent from their sequence.

Information about Gene Regulation. Although DNA sequence does not immediately reveal information about gene regulation, sequencing projects have made possible the large-scale identification of sites of regulatory protein action. This ability comes from comparison of sequences near coding regions in somewhat diverged organisms (*Caenorhabditis elegans* and *Caenorhabditis bergerac*, *Drosophila melanogaster* and *Drosophila virilis*, *Mus musculus* and *Fugu rubripes*). Nucleotides conserved in noncoding regions between these pairs of organisms identify functional sites, typically, sites of action of regulatory proteins, and these sites typically have conserved function when used to regulate transgenes (see for example Venkatesh et al., 1997). Although it is useful to identify regulatory sites from sequence alone, a comprehensive understanding of how those sites work together with their cognate proteins to regulate gene expression will have to await other technologies.

Information about Phylogeny and Evolution. One of the most marvelous attributes of organismic DNA sequences is that they show the changes that have led to speciation and existing phylogeny. By making evolutionary change so starkly visible, the ability to track evolution by its DNA trail makes the concept far less abstract. DNA sequencing has already revealed a number of genomic rearrangements, including duplication events and syntenic rearrangements for many phyla. In one significant example, it is now possible to speculate that vertebrate genomes may represent a quadruplication of an ancestral metazoan genome that also gave rise to worms and flies (Spring, 1997). Analysis of DNA sequence has also

led to readjustments in phylogenies that were based on morphological or paleontological information. For example, early on, working with 18S ribosomal sequences, Lake and his coworkers suspected that lophophorates are protostomes (Halanych et al., 1995), and the accumulation of DNA sequence data since then has confirmed that insight and led to the current picture (Aguinaldo et al., 1997; de Rosa et al., 1999) in which all bilateral animals are either deuterostomes, lophotrochozoans, or ecdysozoans. More recently, Venkatesh et al. (1999) have used spliceosomal introns, some of which hopped in rather recently, as markers to define clades in the ray-finned fishes, and, in the process, helped fine tune this phylogeny. Finally, DNA sequence demonstrates numerous individual instances of horizontal gene transfer among prokaryotic species (Jain et al., 1999). In all, the next few decades should result in a much more complete picture of the natural history of living species.

Genomic Information on the Way

DNA Sequencing and Mapping. After information on the sequence of various species, the Next Big Thing coming from DNA mapping and sequencing efforts is information about the genetic polymorphism within species. Different people are said to differ at 1 in 900 base pairs (Barbujani et al., 1997). The US government now officially (see <http://www.nhgri.nih.gov/DIR/VIP/glossary>) defines polymorphisms as sets of alleles in which the rare form allele is present in more than 2% of the population, with any rarer allele defined as a mutation. Given the genetic diversity of the US population, this definition means that many alleles that we call polymorphisms are rarer in other populations and would count as mutations, for example, in Slovenia or Finland. In fact, polymorphism data has already allowed reconstruction of the migration of different human populations during prehistory (Jin et al., 1999) and is beginning to shed light on other population-related historical and anthropological issues (for a review, see Owens and King, 1999).

For polymorphisms in coding sequences, regulatory regions, and splice junctions where location suggests that they have functional consequences, identification of the polymorphism will often yield more questions than answers: although primitive tools (antisense nucleic acids, nucleic acid aptamers, ribozymes, and peptide aptamers) for probing the consequences of functional changes in proteins exist (see below), there is now a dearth of methods to give insight into the functional consequences of variation in noncoding sequences. It now seems likely that most polymorphisms outside of coding and regulatory regions will lack functional consequence but will be useful as mapping markers for nearby functionally important alleles. Whether functional or not, many polymorphisms will be closely linked to inherited traits, and identification of them (see below) will thus trigger a flood of more or less accurate predictions about genetic predispositions and outcomes without immediately suggesting means to affect those outcomes.

Other DNA Genomics. In short order, it is likely that comparative genome hybridization (CGH) will give insight into the natural history of human cancer (Kallioniemi et al., 1992). CGH can be performed, for example, by making an ordered array of human DNA from intervals

along the genome, then comparing hybridization of differently labeled normal and tumor DNA to that array. Application of this technique in clinical settings should in a few years result in a definitive taxonomy of larger-scale differences between DNA in different tumor types and wild-type DNA. Such work should thus, at a minimum, lead to completion of the set of human recessive oncogenes for which loss of heterozygosity (LOH) can lead to cancer development, and to identification of LOHs and other DNA alterations that are associated with different clinical outcomes.

The next few years will also see a great deal of information gained from more advanced DNA analysis tools. For example, there is increasing evidence that covariation of mutations in the coding sequences of unlinked genes during phylogeny is a sign that those proteins may interact (Horn et al., 1998). More recently, several workers have shown, for proteins involved in metabolism, that if there exists an organism in which two coding genes are fused, that those two gene products may interact in other organisms in which the genes are not fused (Enright et al., 1999; Marcotte et al., 1999a, 1999b). There are likely to be other kinds of information to be gleaned from DNA sequence alone. Because most computational methods are easier than corresponding wet-lab methods, biologists will push the computers as hard as they can.

RNA Genomics. A few years ago, it became technically possible to detect the level of expression of large numbers of mRNAs in a sample at once, by hybridizing to surface immobilized arrays of nucleic acids. The first of these developed used arrays of photolithographically synthesized oligonucleotides (Lockhart et al., 1996). The use of these arrays to monitor gene expression got off to a slow start, largely because the technology was developed commercially and was expensive, which speeded the development of a number of non-array-based methods to survey mRNAs that are not widely used in research labs and that I will not review here. More recently, cheaper and nonproprietary means to produce arrays from PCR products were developed (DeRisi et al., 1997). Propagation of this later technology for making nucleotide arrays has greatly increased the speed at which gene expression monitoring methods are being adopted and producing important results.

One of the first fruits of these experiments has been the identification of new genes involved in known processes. For example, in *S. cerevisiae*, a number of proteins form a protein degrading complex, the anaphase promoting complex or APC, that is needed for cell cycle progression. Gene expression monitoring experiments reveal that mRNAs encoding this complex are expressed during sporulation. This result indicates that protein degradation handled by this complex may be needed for progression through sporulation as well (Chu et al., 1998). Similarly, in human fibroblasts, genes involved in wound healing are expressed when starved fibroblasts are induced to proliferate by serum. This fact is at least consistent with the idea that wound healing is a normal function of proliferating fibroblasts and has caused many investigators studying the response to serum stimulation to see their results in the light of this biology (Iyer et al., 1999).

The promise of gene expression monitoring is not

confined to identification of new genes involved in known processes. For example, by comparing whole genome mRNA expression patterns among a number of mutant strains of *S. cerevisiae*, Holstege et al. (1998) suggested a new function for Srb5 and Srb10, two proteins associated with the transcription apparatus. These results suggest that Srb5 and Srb10 are involved, respectively, in yeast mating and nutrient sufficiency. This study thus suggested new regulatory relationships in which ancillary transcription factors were direct targets for signal transduction pathways.

Because it is possible to isolate different functionally important mRNA subpopulations, gene expression monitoring can give insight into different aspects of cell biology. For example, isolation of cellularly encoded capped mRNAs after poliovirus infection has allowed researchers to identify cellular genes that may be needed for productive viral infection (Johannes et al., 1999). Isolation of mRNAs associated with polysomes makes it possible for researchers to observe expression of genes whose mRNAs are translated. Isolation of mRNAs associated with the endoplasmic reticulum should allow analysis of expression of genes whose products are expressed on the membrane or secreted. Isolation of newly synthesized mRNAs should allow crude determination of rates of synthesis. Because of the large number of techniques that can define different functional mRNA classes, the utility of gene expression monitoring for addressing cell biology is likely to increase.

Protein Genomics. Three ongoing kinds of experiments are now systematically producing information about genome encoded proteins. First, large-scale surveys of protein content in samples using two-dimensional gels (O'Farrell, 1975), coupled, typically, with identification of the protein in individual spots by mass spectrometric determination of molecular weight and sequence, have resulted in some knowledge of the proteins expressed under different conditions in *S. cerevisiae* (Shevchenko et al., 1996) and *Hemophilus influenzae* (Link et al., 1997). A good deal more such information for other organisms and cell types has been collected commercially and is not in the public domain.

Second, again in *S. cerevisiae*, the facility with which genes can be replaced by derivative genes and altered by insertional mutagenesis has made possible an effort to determine the subcellular localization of a significant portion of the proteins expressed in this organism. In these experiments (Ross-MacDonald et al., 1999), proteins encoded by the yeast genome are systematically tagged with an epitope, and the localization of the epitope is monitored by immunofluorescent light microscopy.

Third, two-hybrid experiments (Fields and Song, 1989) can be scaled up by interaction mating to allow the mass testing of interactions among binary protein pairs (Finley and Brent, 1994). This tactic is being used to survey the interactions among the protein products for which the equilibrium dissociation constant of the binary interactions is tighter than about $<10^{-6}$ M (Estojak et al., 1995). Information from such surveys, including interactions among mammalian and yeast proteins, is now stored in searchable databases of various types, and it seems likely that data from a large survey of interactions among most of the proteins encoded by *S. cerevisiae*

will be available within a year. Again, a great deal more information from this approach is not in the public domain.

Genomic Information on the Horizon

The kinds of systematically generated information noted above are only a start. The next 10 years will see a proliferation of new types of genomic data.

First, a large-scale effort to solve protein structures is now beginning. This structural genomics initiative (in the US, the Protein Structure Initiative) is still taking shape. Here, a major issue is how many prototype protein structures must be solved in order to enable accurate modeling by homology of proteins that are related to the prototype by sequence. The ability to make a good guess at the structure of most proteins will have massive long-term returns. These include a greatly facilitated and broadened ability to design structural classes of small chemical compounds that can inhibit the action of given classes of proteins and a scientifically enlightening correlation of different protein structures and substructures with their function, at whatever level their function can be defined—enzymatic function, function at the level of protein interaction, etc.

Second, for organisms whose protein complements have been sequenced, systematic precipitation of individual affinity-tagged bait proteins from cellular extracts allows identification of the other proteins that complex with and are coprecipitated by the bait. The most versatile means of identifying those coprecipitated proteins is protein mass spectrometry. Knowledge of genome sequence enables prediction of their mass, and comparison of predicted mass with that measured by mass spectrometry often allows identification of the coprecipitated proteins (Grant et al., 1998). This coprecipitation/mass spectrometry approach is seemingly better suited to determining the membership of high affinity, multisubunit protein complexes than two-hybrid approaches, and information from it is likely to flow in increasing amounts in years to come. The coprecipitation approach can also enable exploratory biochemistry: recently, Martzen et al. (1999) made use of a collection of yeast for which individual proteins were Gst tagged. They precipitated the Gst-tagged baits, screened the coprecipitates for new biochemical activities, and found them, including a hitherto unknown phosphodiesterase, and a cytochrome C methylase.

Third, in certain organisms, it is possible to generate large-scale collections of uniquely identifiable loss-of-function mutations. Such manipulations are relatively easy in yeast and, recently, have been made possible by genome-scale insertion mutagenesis in embryonic stem cells from *Mus musculus* (Yoshida et al., 1995; Hicks et al., 1997; Zambrowicz et al., 1998). This ability to generate such mutant collections has spawned large-scale efforts to assign function to the wild-type genes based on the loss-of-function phenotypes. Generally speaking, the difficulty in deriving information from the study of such mutants lies in guessing function from the mutant phenotype. In yeast, it is easy to score a few phenotypes, but although some of these (such as nutritional auxotrophy) can point directly to the function of a wild-type gene, others (such as cold sensitivity) do not. In mouse, by contrast, the problem is not the paucity of phenotypes, but rather their great number, so that

the effects of a loss of function can be manifested in any of hundreds of different cell types at any stage throughout development. Nor are these mouse phenotypes necessarily easy to score. Adequate description of the phenotype of a given mutation may require a great deal of knowledge of mouse biology, anatomy, histology, and pathology. Moreover, in mouse, as in yeast, some phenotypes point directly to function, but many others do not.

Two approaches offer some hope in overcoming the limitations. First, in *C. elegans*, a great deal of progress has been made in systematizing the task of scoring mutant phenotypes, so that at least this part of the work can now be performed by relatively untrained people. For the mouse, correlation of molecular phenotypes, for example, changes in gene expression, with organismic ones should allow a similar systematization. Second, and perhaps more originally, there has been a gradual development of genomic tactics to give information about the contribution genes make to organismic fitness and survival under different selective conditions. These schemes, until now only demonstrated for yeast, use large mutant collections in which the individual members can be distinguished by nucleic acid techniques: by anchored PCR using a DNA tag such as a transposon (Smith et al., 1996; Ross-MacDonald et al., 1999) or by a coded nucleotide sequence specific to each mutant gene (Winzeler et al., 1999). One collection is haploid cells with loss-of-function mutations. Another is diploid cells that are haploid for different loci. When such collections are subjected to growth under some selective condition (including, for example, nutritional deprivation, temperature shift, overexpression of a gene, or presence of a chemical compound), census of the members of the collection present in the selected population at different times reveals what genes are essential for growth under that condition, which are useful, and which are irrelevant (Giaever et al., 1999; Winzeler et al., 1999). These approaches are likely to be extended to other organisms, at least to worms.

Fourth, new technologies under development will, if successful, lead to systematic production of new kinds of functional information. For example, in an approach pioneered by Tsien and coworkers (see Heim and Tsien, 1996) fluorescence resonance energy transfer (FRET) can reveal protein interactions in living cells. In such experiments, one interacting protein is fused to a GFP-derived FRET donor and the other to a GFP-derived FRET acceptor. Interaction between the proteins is monitored by fluorescence microscopy and/or light spectroscopy; if FRET occurs, the donor fluorescence is reduced and the acceptor fluorescence is increased. Although implementation of this idea on a genome-wide scale would require surmounting a number of technical issues (Mendelsohn and Brent, 1999), its use has already revealed mitochondrial interactions between Bax and Bcl-2 (Mahajan et al., 1998) and allowed real time assay of interaction among components of the nuclear pore apparatus involved in protein import (Damelin and Silver, 2000). Still more sophisticated ideas are under development. These include methods to characterize the static and dynamic complement of cell proteins, protein association states, protein modification states, and proteins bound to regulatory sites on DNA. Those methods that

give useful information relatively cheaply will be scaled up. The right concatenation of methods may allow understanding of the flow of information through the cell and how that information is used to regulate cell function.

Using Genomic Data to Gain Biological Understanding

The difficulty in using genomic data to quickly generate biological knowledge has led many contemporary biologists to a degree of cynicism about the genomic enterprise. Genomic data have common characteristics that would seem to justify a measure of cynicism. First, genomic data of all types are typically lower value adding than those obtained from classical ad hoc experimental approaches; they typically do not immediately address the questions about regulation, mechanism, and decision making that are of greatest interest to contemporary biologists. Second, due to limitations of the individual methods and pressure to mass produce the information, all forms of genomic data are prone to errors. Even sequence data, which are subject to rigorous quality control, become encrusted with half-truths and outright falsehoods at the moment they are annotated. Third, with the exception of sequence data, both the computational methods and the underlying logic used to make inferences from genomic data are in a primitive state characteristic of an observational stage of science. Consider gene expression data. The first clustering methods (Weinstein et al., 1994, 1997) led to faster (Eisen et al., 1998) and more sophisticated means (Tamayo et al., 1999) to group individual observations so that one can think about them. Analytical methods now under development often rely on a single, simple concept: "something's different over here; perhaps it's important." And some of the most sophisticated observation studies to date have relied on logical tactics that, although they work, are best described by the names of their corresponding fallacies: a gene whose transcription behavior resembles that of a known gene may function in the same process as the known gene (guilt by association); a gene whose transcription is induced before transcription of a group of another genes may regulate transcription of that group of genes ("post hoc, ergo propter hoc" or "after this, therefore because of this"). Logical and computational techniques to make sense of protein interaction data are even less well developed. For these reasons, information gained from genomic data rarely rises to the level of the conclusions that biologists prefer but often hovers at the level of suggestion, indication, inference, or testable hypothesis.

One obvious path toward raising the biological value of genomic information is to combine different genomic data types. For example, the conjunction of a finding that two proteins can interact with a finding that mRNAs encoding those proteins are expressed in the same cell at the same time strengthens the idea that the two proteins do interact. Computational combination of different data types to make stronger inferences is a difficult problem but one whose solution will offer significant near-term benefits (see below).

Genomic Experimentation

A good deal of genomic information is produced systematically in facilities and by work teams specialized for

that purpose. As mentioned, it may be possible to combine different kinds of this systematically generated information to get closer to an understanding of function. However, at the moment, we are not in that promised land, and most of the strong conclusions will continue to come from directed experimentation. The problem, of course, is that directed experimentation is difficult, typically requiring bright researchers who have trained for years in contemporary biology and who are expert in the system in which the experiments are performed. There are not enough such researchers to use existing tactics to solve existing biological problems in reasonable time. Thus, adequate progress in increasing biological understanding will depend on the development of better techniques.

It is possible to list the ideal attributes of such genomic experimental techniques (Table 1). First, of course, they should be cheap, accessible, nonproprietary, and undertakeable ad hoc by individuals and small research groups seeking answers to specific questions. Second, they should be generic, in that they should not depend on special properties of the organism that contributed the genes. Third, and in the same vein, they should not require difficult genetics, breeding, careful observation, or insight to be performed—in fact, not to put too fine a point on it, the best future experimental methods to help understand biological function should not require that the experimenter be highly skilled. The fact that such genomic experimental techniques will not require great skill will mean that it will be possible to systematize them, so that if there is a clear need to perform them upon entire genomic complements, they can be scaled up and industrialized.

A number of methods to probe the function of individual gene products seem to fit this description. In each case, these use agents that can be obtained or isolated with relatively little effort and that promise insight into the function of single genes and small gene collections. At present, these agents are all affinity reagents that can be isolated from combinatorial libraries, including antisense nucleic acids (Izant and Weintraub, 1984; Melton, 1985), nucleic acid aptamers (Ellington and Szostak, 1990), peptide aptamers (Colas et al., 1996), and antibodies from recombinant DNA libraries (Ward et al., 1989). In the future, these agents may include small molecules that interfere with the function of individual gene products. It is not clear that collections of such compounds can be made, but, if they can, their manufacture will be beyond the abilities of individual researchers, who would instead identify them and obtain them from centralized resources.

All of these approaches share the characteristic that the perturbing agent is dominant, and so their introduction into diploid cells leads to the loss of the function of the product of both copies of the gene. However, at least for peptide aptamers, which display a conformationally constrained variable region from a rigid scaffold, it is possible to isolate molecules that recognize one allelic form of a protein but not the other (Xu et al., 1997). Allele-specific interfering agents are likely to give insight into the diversity of gene function caused by polymorphism in protein coding regions. Very recently, it has also proved possible to use these agents in forward-"genetic" schemes: in yeast, peptide aptamers have

Table 1. Genomic Methods

Production	Ad Hoc Experimental
DNA sequencing	Reverse genetics
Polymorphism detection RFLPs, SNPs, etc.	Targeted mutagenesis with DNA constructs
Gene expression monitoring	Reverse "genetics"
Two-hybrid mating	Targeted interference with protein function using specific intracellular antibodies, antisense nucleic acids, ribozymes, nucleic acid aptamers, and peptide aptamers
Coprecipitation with affinity-tagged bait	Forward "genetics"
Large scale visualization of subcellular localization	Identification of peptide aptamers or other agents that cause a phenotype, followed by identification of proteins and protein interactions targeted by those agents
Genome-wide insertional mutagenesis	
Loss of function, others	

Comparison of genomic "production" methods and genomic ad hoc experimental methods. If the cost to perform a given information producing technique, for example, gene expression monitoring, drops, the information may move to the ad hoc category; and be performed to test a specific idea, conversely, if it becomes useful to generate a certain type of information or reagent on a genome-wide scale (for example, should it be useful to generate affinity reagents against all encoded proteins), a technique may move to the production category.

been isolated to cause a phenotype and then used as baits to identify proteins and protein interactions that cause the phenotype (Geyer et al., 1999; Norman et al., 1999).

Consequences of Genomic Research Changes in Health and Agriculture

Biological understanding affects the human condition via two systems that develop and make use of it: the medical-pharmaceutical system and agriculture. To date, genomic research has most heavily influenced the discovery research sector of the pharmaceutical industry. Here, in many companies, combinations of genomic information, including sequence data, gene expression data, and protein interaction data, are used routinely to identify proteins that might make good targets for drug discovery efforts. As soon as they are robust enough to work reliably in industrial labs, other genomic methods, including selective growth of pools of mutants, large-scale gene knockouts in mouse, and aptamer-guided target selection will be used to identify and validate drug targets.

It will not be long before genomic methods begin to affect downstream aspects of pharmaceutical research and development. Correlation of different patterns of gene expression with different pharmacological and toxicological outcomes will allow gene expression data to supplement and perhaps substitute for more conventional toxicological studies on animals. Analysis of human polymorphisms will allow delineation of patient subpopulations during clinical trials. For example, people who carry certain allelic variants of cytochrome P450 CYP2D6 do not convert codeine into morphine and do not benefit from the drug, while people with amplifications in CYP2D6 metabolize codeine so well that standard doses are ineffective (reviewed in Eichelbaum and Evert, 1996). Similar polymorphism-based stratification of patients will help identify subgroups that benefit from or are adversely affected by tested drugs. Similarly, it is likely that gene expression monitoring will be developed to the point that changes in patterns of expression of particular genes, in patients with particular genetic makeup, will provide early warnings of therapeutic or toxicological outcomes. Stratification of patient populations and use of faster techniques, such as gene expression monitoring, to forecast outcomes should shrink the number of patients and the length of time for drug trials.

It is difficult to exaggerate the benefits that should flow from these changes in drug development. Use of genomic techniques should place the classification of drug side effects and adverse reactions on a more rational and mechanistic basis and eventually allow patients to receive drugs appropriate for their genotype. This development should permit the development of prophylactic drugs whose preventive effects are sufficiently beneficial and whose side effects are sufficiently mild to allow them to be taken for decades. Perhaps as important in the long term, by decreasing the time and number of subjects needed for clinical studies, the use of genomic information should greatly decrease the cost of developing new drugs. A radical decrease in cost would permit drugs to be developed for smaller markets, including particular genetic subpopulations and patients with diseases endemic to poorer countries.

In clinical medicine, the classification of patient populations by genotype will allow careful examination of which therapies are effective and which are not on particular subtypes. Such subdivision will also allow simple longitudinal studies that identify genetic markers that confer predisposition to many diseases. These developments make it likely that the decades ahead will be an anxious interlude in which ability to predict predisposition will outstrip mechanistic understanding of the conditions deep enough to guide development of therapeutic pharmaceuticals. Diagnostic and prognostic ability without effective drug therapy is difficult for patients, physicians, and the health care system. For example, at present the most effective therapeutic response to the knowledge that a patient carries a predisposing mutation in BRCA1 is prophylactic mastectomy (Hartmann et al., 1999).

By contrast with drug development and health care, which intervene directly in human biology, the revolutionary impact of genomic technologies on agriculture will likely encounter fewer barriers. In fact, good polymorphism and sequence data for many economically important plant and animal species already exists (in private hands). Such genetic, non-mechanism-based, correlative data are even more immediately useful in agriculturally important species than in medicine, because breeding allows the correlation of genetic markers with favorable traits in offspring of a cross, and the use of genetic methods to identify and manipulate subtle,

multigenic traits is well accepted (quantitative trait localization, or QTL [Sax, 1923; Edwards et al., 1987]). Moreover, it is easier to test the utility of an experimental intervention in plants and animals than it is in people, because many of those manipulations can be based, not on intervention with specific small molecule chemicals, but on changes in the organism's DNA. Although agricultural genetic manipulations will inevitably be regulated by governments, the ease of suggesting and making desired manipulations should ensure that, by the middle of this century, most of the plants in any cultivated landscape will carry genetic alterations guided by genomic research.

Changes to the Biological Enterprise

Deemphasis of Hypothesis-Driven Research. The idea that valuable knowledge can be gained from the systematic production of simple kinds of biological information has lessened the emphasis on strictly hypothesis-driven research. The idea that science proceeds by the formulation and careful testing of hypotheses can be traced at least to Francis Bacon, more than 350 years ago (Bacon, 1620). Although most contemporary biologists believe in this idea, most also have personal experience with successful inquiries that were better described as observation-, obsession-, engineering-, or "what if"-driven, in which testing of hypothesis played a small part. In fact, in a great deal of biology, the Baconian picture is best applicable to troubleshooting (the same reason it is used, for example, by computer repair technicians and automobile mechanics). Certainly, much ongoing genomic research is better described as observational than experimental, and, as mentioned, its findings are sometimes better described as inferences or hypotheses for further testing rather than rising to the level of the conclusions that biologists are trained to seek. Moreover, many hypotheses resulting from genomic observation will never be tested because scientific standards have changed. For example, in 1988, the finding that a protein contained a homeobox might suggest that it bound DNA and regulated gene expression, but that suggestion would probably have been tested experimentally. In 2000, we would probably accept that claim without further experiment.

In biomedical research, the emphasis on hypothesis is one consequence of the concordat between biologists and physicians that enabled the US NIH to support basic research. The bargain has served medicine well—it vindicated the idea that basic research leads to better medicine, and medical researchers, as part of their training, were indoctrinated with the idea that they should frame hypotheses and test them by controlled experiments. The fact that many of the hypotheses were stereotypical (e.g., transcription of genes in the kidney may be controlled by transcription regulatory proteins present in the kidney) made possible a rapid expansion of mechanism-based biomedical research, particularly when molecular biological experimental techniques became more accessible (Maniatis et al., 1982; Ausubel et al., 1987). However, this agreement may not have served biology as well as it served medicine. As insistence on the hypothesis-driven heuristic became enshrined and championed (often by reviewers and study section members), the biological community became less tolerant of other

kinds of experiments and experimenters. Young scientists are sensitive to perceived hypocrisy (Lewis, 1925), and insistence on a framework visibly at variance with the way some research is done may even have driven some of them into other lines of work.

The fact that genomics is forcing biologists to remember that not all research is driven by hypotheses is not to say that biologists can stop thinking. At least in biology, it is fallacy to think that accumulation of data alone can lead to insight. However, there are people who enjoy collecting information. If, in the limit, there were to arise obsessive-compulsive types whose highest goal was to accumulate data, biology as a whole might not lose so long as other people learned to analyze the data and test the analysis by experiment.

Enforced Dialog with Computer Scientists. That genomic information comes computer ready, that individual observations are typically not typically very valuable, and that there are so very many individual observations all have led to the utter dependence of genomic research on computer science. This reliance will only increase. Each new type of genomic information is different and requires different structures to store it, different defined relationships among its data elements, and different tools to access and understand it. This level of computer work is often tedious for the biologist. Many of the abstractions needed to understand the computational treatment of genomic information come from the realm of relational databases (Date, 1994). These concepts, including the reduction of a universe of knowledge to defined types of entities, the enumeration of those entities in tables, and the connection of those entities by defined relations, date only from the 1970s (Codd, 1970). Many people find it difficult to think along these lines. However, although it is difficult, it is urgent that high-level biologists establish a more effective dialog with those computer scientists versed in the classification and retrieval of information and for these biologists to participate in serious discussions to establish the frameworks (data entities, allowed relationships, meaningful relationships) for each new data type. Because newer data types will continue to be developed, this sort of database-intensive, bioinformatic work will need to continue for decades. However, beyond this database bioinformatics, the challenges presented by integration of different genomic data types with one another and with natural language data, and by construction of higher-level computable representations of biological knowledge, dictate that biologists will also need to work with outstanding computer scientists outside of the database field.

Increased Class Stratification and Division of Labor. Access to the newest results, the most talented colleagues, the most modern methods, and the best reagents and equipment has always been uneven. Genomics is accentuating this inequality in at least two ways. First, a great deal of genomic data is produced by corporations and is slow to find its way into the public domain. As this article goes to press, the sequence of *Drosophila melanogaster* has been available to corporate partners of Celera for more than a month; it will be released to the public sometime during 2000. Similarly, one Incyte database, accessed by subscription, has consistently

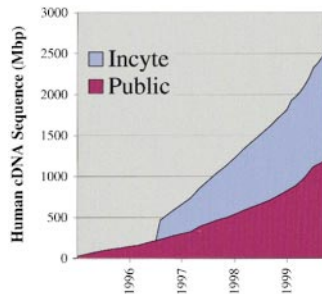


Figure 1. Private and Public Genome Data

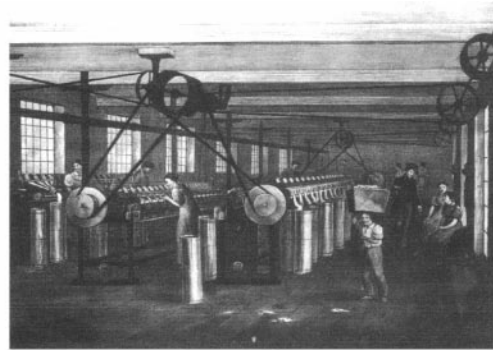
Figure shows total human cDNA sequence (in millions of base pairs) in a private database (LifeSeq, Incyte) and a public one (dBest). Subscribers to the private database can, of course, also access the public data. The quality of the private sequence is said to be higher than that of the public (LifeSeq statistics courtesy of Incyte, Inc.).

contained more human cDNA sequence than the equivalent public databases (Figure 1). As new types of genomic information are developed, corporations will arise to produce it, and the business plans of those corporations will often involve selling the information only to a few large corporate partners. Second, computational tools that allow the best use of genomic information will often be first perfected by corporations whose business models require them to supply those tools to corporate partners. Although it is in the long-term interest of both the public and private sectors that the gap in access and ability to use genomic information not grow wider, the construction of public policies and commercial business models that favor this outcome is now an unsolved problem.

Along with increased class stratification has come an increased division of labor. Even 10 years ago, most research groups generated most of their own data, performed their own analysis of it, and, based on that analysis, performed new experiments to generate more data. Genomic research has disrupted that cycle by creating a class of specialized information producers and a set of specialized industrial activities collectively referred to as production (Figure 2). As with previous industrial transformations, production workers have become less autonomous, and those who own or control the means of production have become more powerful. The industrialization of production of other types of biological data may hasten the rise of a new type of researcher, the "analyst," who makes use of the information to make suggestions, but who is not particularly insightful or skilled about devising tests for them. It may also sharpen the function of experimenters by defining their role as adding value to the systematically produced data and inferences from it by focused experiments.

Stimulation of Design-Based Biological Engineering. Biological knowledge affects the human condition via medical and industrial technologies. At the core, both technologies are engineering endeavors, in that both involve the informed and imaginative use of existing knowledge and techniques to produce specified outcomes (for medicine, diminution of suffering; for industry, outcomes that serve some market). Despite that similarity, the education of engineers differs from that

A



B



Figure 2. Production

(A) Cotton thread production. Drawing carded cotton into yarn, England, presumably Lancashire, circa 1820 (courtesy of Yale University Gallery).

(B) Human cDNA sequence production. Sequencing ESTs, Palo Alto California, circa 1995 (courtesy of Incyte, Inc.).

of physicians and industrial biological scientists. As undergraduates, engineers are encouraged to value achievement of technically difficult goals for their own sake. For example, consider the fabled internet server in 1975 that revealed the number of cans in the Coke machine in a Carnegie Mellon University basement (Kelly, 1993). There is at present no understanding of this hacker mindset, the joy in engineering for its own sake, in the biological community.

The genome projects, with their promise of complete parts lists, have caused would-be engineers to start turning up in biology labs. It is unclear whether the first products of these cellular hackers, typically recreations of cellular clocks, switches, oscillators, etc., will themselves have any immediate positive impact on biology (see Elowitz and Liebler, 2000). However, the engineers who build these devices will affect biology profoundly. By doctrine and training, modern engineers simulate devices whenever possible before they build prototypes. For example, nowadays, almost all digital and analog circuits are first designed and tested using a particular, decades-in-the-making-and-improvement simulation program known as SPICE (McCalla, 1988). Even absent stimulus from biologists, this drive to perform engineering with a rigorous design component will spur development of biological simulations and the collection of information to populate them.

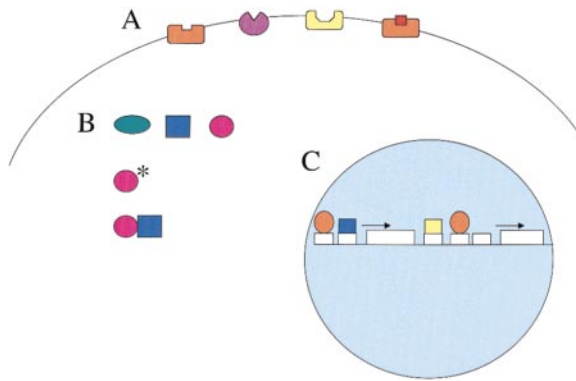


Figure 3. High-Value Functional Information

(A) The identity of receptor proteins expressed on the membrane, and whether they are bound by ligand or not.
(B) The identity of proteins expressed in the cytoplasm, their modification state, and the identity of proteins they are complexed with.
(C) Sites in gene regulatory regions, and the identity of proteins bound to them.

Longer-Term Challenges for Genomic Biology *Developing New Devices and Techniques*

Observational genomic techniques such as gene expression monitoring or protein interaction surveys are often compared to observational scientific instruments such as telescopes and microscopes. To make this comparison is to ask whether the data acquired by existing genomic methods is as scientifically important as the data acquired by astronomers and microscopists. As mentioned above, as biologists discover better ways to use the new information, this question will seem less compelling. Meanwhile, the comparison does have the virtue of making the genomic methods look inadequate and does stimulate thinking about the sorts of observations future genomic biologists could make that would be of more immediate biological importance. Note, when we are discussing observational science, that observations can be made with miniature instruments, with laboratory instruments, with observatories, or with production lines, but that the capital cost determines the number of different observations and experiments one can do and thus probably determines how rapidly knowledge can advance.

With this comparison in mind, let us imagine three types of devices, or scopes, that would provide information about cells that biologists would deem important (Figure 3). The membrane scope would reveal which receptors were expressed on a cell membrane or not and which were bound by ligand. The cytoplasm scope would reveal the presence, absence, abundance, modification state, and association state (if in a protein complex, the identity of the partners) of a selected set of several hundred regulatory proteins. The DNA scope would allow scientists to observe which regulatory sites in a cell's DNA were bound and by what regulatory proteins. The sooner we start imagining how to systematically generate this information, the greater the chance we will be able obtain it sometime during this decade.

Coping with Weakly Penetrant and Modifier Genes

As articulated by Morgan, Sturdevant, Müller, and their students, the forward transmission genetic approach to

understanding biological processes involves the isolation and collection of mutants that show specific phenotypic alterations, the identification of the genes that bear those mutations, and the subsequent study of those genes and their products. This approach was so powerful that investigators readily accepted the limitation that mutations that were lethal to the organism before the mutant phenotype could be scored would not show up in their mutant screens. By the 1960s, at the latest, the doctrinal cloud surrounding this classical genetic approach had added another element, which was that mutants that displayed weak phenotypes or that conferred inconsistent phenotypes in downstream genetic manipulations should not be included in the collection under study. By excluding weak and inconsistent effects (or, as its practitioners sometimes said, by letting the biology indicate the important genes and gene products), this heuristic has been instrumental in helping biologists focus on the genes that are most important.

However, the genome projects are showing us more genes than we knew about before, and there is no reason to think that all of them are responsible for strong, consistent effects. In fact, there is at least some evidence to the contrary. For example, in *C. elegans*, the strongest mutations in a ρ family GTPase (Zipkin et al., 1997), in some genes that affect axon guidance (Peckol et al., 1999), and in some genes that affect formation of dauer larvae (Thomas et al., 1993) are incompletely penetrant. For the dauer mutations, incomplete penetrance is due to the fact that there are genes with redundant functions, that is, the mutations affect pathways for which there exist at least partly parallel pathways. However, for others, the cause is unknown. Similarly, the world of plant and animal breeding has dealt from the beginning with genes whose effects, for example, on size or yield, may be consistent but are subtle and quantitative. For both incompletely penetrant and weak genes, isolation of and study of effects in individual mutant organisms is difficult, and the best present approaches to their study rely on techniques such as QTL (Sax, 1923; Edwards et al., 1987; Lander and Botstein, 1989). These techniques require analysis of fairly large numbers of mutant or hybrid organisms and quantitation of their effects on the population. For traits of sufficient interest (say, susceptibility to carcinogens), these methods will surely be applied to appropriate model organisms and will lead to identification of many weaker genes. These methods will likely be supplemented by functional studies. Here, approaches that create populations of gene-tagged mutants, subject these populations to selection, and monitor the contribution of individual genes to the fitness of the population over time (Smith et al., 1996; Ross-MacDonald et al., 1999; Winzler et al. 1999) suggest a way forward. In organisms and cells in which creation of populations of tagged mutant organisms is difficult or impossible, use of appropriately tagged ersatz genetic agents such as peptide aptamers may provide an analogous path to identifying incompletely penetrant and modifier genes. Because it will be possible to control the temporal and spatial expression of such agents, this approach would allow identification of genes that affect specific processes but that are also necessary for the survival of the cell or organism.

Moving from Databases to Biological Information Systems

To be maximally useful to biology, different types of genomic information will need to be better integrated and will need to be more easily combinable with the qualitative, narrative information now housed in the biological literature. The problems in getting to this point seem difficult but solvable. At the moment, there is, to put it mildly, a dearth of general methods to integrate different types of data residing in different databases. The standard notional solutions to the integration problem are either to put all the information into a single, large database or to persuade groups that maintain different databases to agree to organize them in standard ways with standard internal structures. Neither approach has been conspicuously successful to date. By contrast, individual translation programs that import data from one database into the other do work, but such programs need to be custom written by skilled workers who take into account the internal structures of both databases and who rewrite the translators when those structures change. There is some work on more general solutions to the translation problem, such as the IBM Garlic project, in which each database has added onto it a wrapper that, once written, communicates with a general purpose "middleware" program. Here, the wrapper needs only to be written once, and the user accesses only the middleware and neither sees nor needs to understand the structure of the underlying databases (Carey et al., 1995; Haas et al., 1997). Whatever the means used, integration of genomic data will take leadership and investment and willingness to confront the difficulties caused by errors particular to each data type. Similarly, genomic data need to be integrated in turn with classical biological information resulting from directed experimentation. At present, most such information is represented in natural language narratives that describe individual systems. These narratives typically include a large number of proper nouns (p66, superoxide dismutase), a smaller number of verbs (binds to, dephosphorylates, activates transcription), a smaller number of modifiers (many, some, rapidly), and a small number of prepositional phrases designating location (in the cytoplasm, near the membrane, on the DNA). The restricted vocabulary allows condensed symbolic representations of what is known; one such narrative, which represents the proteins that control the mammalian G1-to-S transition, is shown in Figure 4A (Kohn, 1999).

Fortunately, there has been significant progress from the AI community in devising analytical and computational representations that manipulate this sort of qualitative and semiquantitative knowledge together with quantitative data where that is known (Kuipers, 1994). Although these representations might access databases, they bear more kinship with conventional expert systems. For example, Figure 4B depicts a part of the plumbing in the Cassini Saturn spacecraft. Modern spacecraft sometimes carry computational representations that incorporate qualitative information (connecting pipes), semiquantitative information (a valve can be open, closed, stuck open, or stuck closed), and quantitative information (the rate of fuel flow into the main engine). This representation allows computation of fuel flows in the network when it functions normally and recomputation, leading to rerouted flow, when the network is damaged. Such mixed representations can be

quite ambitious. Consider that, beginning in the 1970s, and spurred in part by the need to provide targeting information to precision guided munitions, the US government funded Geographical Information Systems, or GISs (<http://www.ucgis.org>; Foresman, 1980; Abler, 1987). These systems work because we can accurately describe geographical information with a restricted vocabulary. GISs contain a large number of proper names (place names) that correspond to a smaller number of classes of entities (cities, roads, boundaries) that are connected by a much smaller set of relationships among those entities (direction relations, topological relations, the distance relation; Table 2). Construction of GISs sparked a multibillion dollar industry and development of a large amount of useful software. For example, when one downloads driving directions between two addresses from a service on the web, one is using a program that accesses a GIS and uses the GIS's underlying computational engine to generate the directions.

For a few prokaryotes and subsystems within eukaryotic cells, we are at or near a level of description where we can enumerate key players, molecules and sites, and where the number of classes of relationships among these entities is small (Table 2). The next decade is likely to see the development of representations that allow computation on mainly qualitative biological information for at least some prototype cells. One early payoff of construction of such Biological Information Systems might be an analog of the driving direction application: a program that generates an exhaustive listing of known or predictable qualitative consequences of given perturbations, for example, inhibition of the function of a particular protein phosphatase. Such first order computation on biological information will be more than a teaching tool; for industrial researchers, it may help choose molecules to be targeted for drug discovery efforts, and for twenty-first century clinicians armed with broader pharmacopoeias than contemporary physicians, it may help guide the choice of therapies.

Predicting Biological Outcomes

For centuries, our knowledge of transmission genetics has taught us that important aspects of cell and organismic development are heritable, reproducible, and ordered. For decades, we have known that this ordered development of cells and organisms depends on the reading-out of the program stored in the genome as genome products interact with one another and with the external environment (Stent, 1968, 1985). For decades, many researchers have assumed that many differences in the behavior of genetically identical cells are due to stochastic variations in the number of regulatory molecules present in small numbers (Delbrück, 1945; Spudis and Koshland, 1976). What we know to date gives us confidence that much of the behavior of cells and organisms is determined, and determinable, from their present states.

However, our ability to predict the behavior of biological systems remains poor. This is true whether we speak of predicting the behavior of cells and organisms in response to known changes to their genetic information, to the expressed products of their genomes, or to their external environment. One explanation for this poor predictive ability can be illustrated by a comparison of our understanding of biology with our understanding of orbital mechanics. For a satellite in orbit around the Earth,

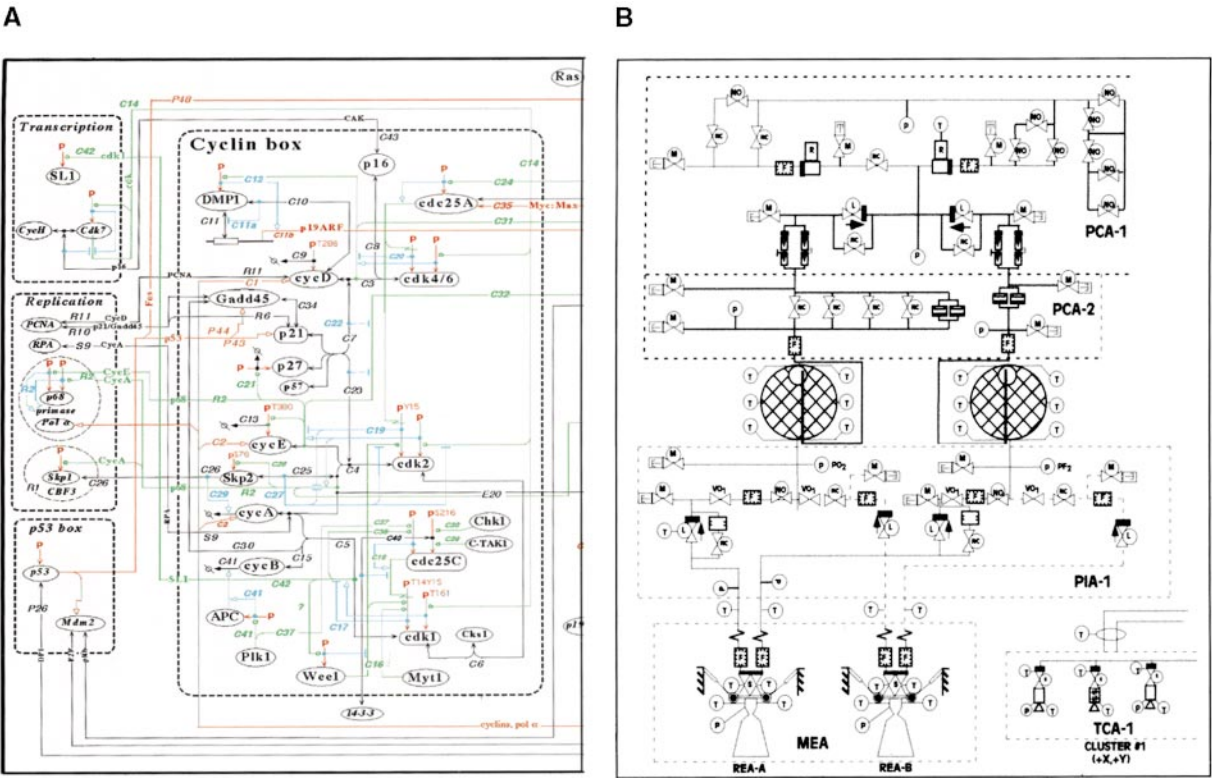


Figure 4. Computable Representations of System Data
(A) Portion of regulatory network governing the mammalian G1-to-S transition. In this representation (Kohn, 1999), proteins and sites are connected by arrows, and the symbols at the heads of the arrows correspond to a limited number of regulatory events, such as protein dimerization, transcription activation, and the like.
(B) Portion of propulsion and attitude control system in the Cassini Saturn mission. In this representation, tanks and engines are connected by pipes, and flow through the pipes is controlled by pumps and valves. Computation on this representation is usually semiquantitative, in that it for example is often sufficient to understand that a valve is open, closed, stuck open, or stuck closed (redrawn from information supplied by Nicola Muscettola, NASA Ames, and Brian Williamson, MIT).

knowledge of its position and its velocity vector at any point in its orbit allows accurate prediction of its future orbit, and knowledge of a change in its velocity vector at any point allows accurate prediction of the resulting change in its orbit. In this example, the predictive power rests on two foundations: the ability to measure the right quantities (here, position and velocity vectors, but not the mass of the satellite and not the rate of rotation of the earth) and on possession of an analytical framework

(here, the calculus) that defines which measurements are important and uses that information to make predictions. At present, biology lacks both the ability to measure the right quantities and the analytical framework that defines which quantities to measure and uses those measurements to make predictions.
Future genomic biology may help us with the measurements, and it will also likely spur development of analytical frameworks. One class of such frameworks

Geographical Information	Biological Information
Names (Berkeley, Berkeley-Oakland line, Interstate 80, Bay Bridge, San Francisco Bay)	Names (p107, Rb, Raf, ATP, Ovalbumin Estrogen Response Element)
Relationships (shares border with, is under, north, right, 5 miles, 11 min)	Verbs (homodimerizes, ubiquitinates, cleaves, represses)
Verbs (go)	Modifiers (strongly, slowly, most, some)
Modifiers (fastest route, under construction)	Locations (in cytoplasm, near plasma membrane, in plasma membrane, in nucleus)
Locations (37.8°N, 122.3°W, 50 m above sea level)	

This shows examples of each class of “words.” Both kinds of information deal with a great many proper names but far fewer significant relationships among those named entities. For geographical information, there are significant relationships that reflect distance, direction, and topology, a few qualitative modifiers, and (when used to generate driving directions) at most a few verbs, but there is a vast number of significantly different spatial locations. For cellular biological information, there are again a few qualitative modifiers, there are many more verbs (although probably less than 100 common ones), and there are only a handful of spatial locations.

is actually borrowed from early computer science. In these, biological systems are represented by Markov chains (Norris, 1997). These chains consist of a finite number of states, defined for example by the presence and absence of different molecules. The probability that a current state makes a transition to one of a number of different possible future states is determined by experiment. Markov representations may be appropriate for systems where the molecular players are well understood, where quantitation can be discrete (e.g., zero, low, and high) and experimental observation is possible; possible examples include λ phage lysogeny (Reichardt, 1975; Maurer et al., 1980; Meyer et al., 1980) and differentiation of adult muscle cells from precursors (Cornelison and Wold, 1997). The advantage of Markov chain representations is that they do not require knowledge of the kinetics with which species are converted into other species. Although creation of strongly predictive simulations of this kind may be quite difficult, construction of them does force biologists to specify key states and measure the transition probabilities and so carries heuristic value. And it is possible, although so far not proven, that quantities derived from these representations, for example, entropies of individual state transition probability distributions or the entropy of the entire chain (Khinchin, 1957), may be useful for describing and comparing systems and might even alert researchers to cases in which the lists of molecular species that define the states are incomplete or incorrect.

Better predictive ability may depend on representations that incorporate kinetic information. The classical frameworks for this are, of course, systems of differential equations that describe the rates at which enumerated species change, via enumerated reactions with enumerated forward and reverse rate constants, from one into another. Although large systems of such equations are difficult to solve analytically, the increase in computer power over the past 40 years has vastly simplified their approximate numerical solution. A variation on this approach, based on work by Gillespie (1977), originally intended for stochastic chemical kinetics, has allowed the numerical solution of differential equations where rate constants are embodied in the increment or decrement of small numbers of molecules over slices of time. Stochastic kinetics is important because it mirrors what actually happens in cells: when the number of regulatory molecules is small, their number in individual cells can vary greatly and cause variation in cell behavior, but when one examines a hundred cells or runs a stochastic simulation a hundred times, one gets an envelope of possible behaviors (McAdams and Arkin, 1997). A more recent framework represents biological systems, for example, cells, as collections of the molecules or stable molecular complexes they contain. Interaction of these discrete entities with one another depends on diffusion, and the probability that individual collisions result in a change to given species is determined by generating a random number and using the number together with a table of precomputed transition probabilities to determine the outcome (Morton-Firth et al., 1999).

Such predictive frameworks will require exhaustive knowledge of molecular species and of rates or rate surrogates: such as numbers of species turning into other species per time interval or collision probabilities

and transition probabilities after collision. Construction of such simulations will need to be tightly coupled to an ongoing program of biological experimentation to collect the input information and test the models. Collection of this information will likely require new, general, systematizable (i.e., genomic) methods. Despite the difficulty of these tasks, their accomplishment will offer significant benefits. As mentioned, simulations will become important design tools for engineers. Simulations will also help biological research and the technologies that apply it. To give one example, they should allow identification of points at which simultaneous small perturbations should have large consequences for system behavior. For medicine and agriculture, such knowledge should translate into ability to intervene with two or more compounds, each of which causes only a small diminution in the activity of its target, to allow achievement of desired effects from lower doses of compounds (fewer side effects, fewer environmental consequences) or from compounds that have not been through so many rounds of chemical modification (faster time to compound).

Conclusion

The wave front of genomic data gathering, experimentation, mechanism-based model building, attempted prediction, and model testing will move beyond cells to systems of cells within organisms, multicellular organisms, and populations of organisms interacting with their environment. The potential impact of this increase in knowledge and capability during the twenty-first century is at least as large as the changes that development of electronics and computation brought to the twentieth. During the next hundred years, genomic discoveries will lead to significant alterations to all economically important organisms and to our own species. Genomic biologists must both do the science and play their part in the political debate that these expanding abilities will provoke.

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